

Pulmonary circulation ,edema and pleural fluids

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Outlines of presentation

- Pulmonary circulation
- Pulmonary Edema
- Pleural Fluid

1. Pulmonary circulation

◆ The lung has two circulations:-

i. **High-pressure, low-flow circulation**

- ✓ supplies **systemic arterial blood** to trachea, *bronchial tree*, supporting tissues of the lung, and outer coats (adventitia) of pulmonary arteries and veins.

ii. **Low-pressure, high-flow circulation**

- ✓ supplies venous blood from all parts of the body to the alveolar capillaries

◆ **Lymphatic**

- ✓ Lymph vessels are present in all the supportive tissues of the lung → right thoracic lymph duct
- ✓ Drain particulate matter entering alveoli and leaking plasma proteins (Prevent pulmonary edema)

1. Pulmonary circulation

PRESSURES IN THE PULMONARY SYSTEM

Pressures in the Right Ventricle

- ✓ systolic **25 mm Hg**, and diastolic pressure 0 to 1 mm Hg

Pressures in the Pulmonary Artery

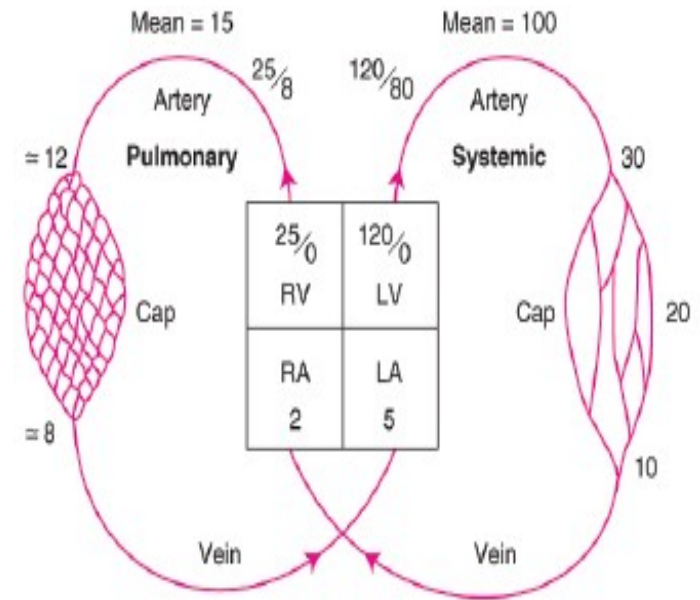
- ✓ systolic pulmonary arterial pressure about **25 mm Hg**
- ✓ diastolic pulmonary arterial pressure is about 8 mm Hg
- ✓ mean pulmonary arterial pressure is 15 mm Hg

Pulmonary Capillary Pressure

- ✓ mean pulmonary capillary pressure about 7 mm Hg

Left Atrial and Pulmonary Venous Pressures

- ✓ MAP: 1 mm Hg - 5 mm Hg (2 mm Hg in the recumbent position)



Pulmonary circulation

Blood flow through the lungs and its distribution

- ◆ The blood flow through the lungs is essentially equal to the cardiac output
- ◆ The factors that control CO also control pulmonary blood flow
- ◆ Decreased alveolar oxygen reduces local alveolar blood flow and regulates pulmonary blood flow distribution (*opposite to the effect observed in systemic vessels*)

Pulmonary circulation

Effect of Hydrostatic Pressure gradients in the lungs on regional pulmonary blood flow

- ◆ Under different normal and pathological lung conditions, one may find any one of three possible zones (patterns) of pulmonary blood flow:

i. **Zone 1: No blood flow**

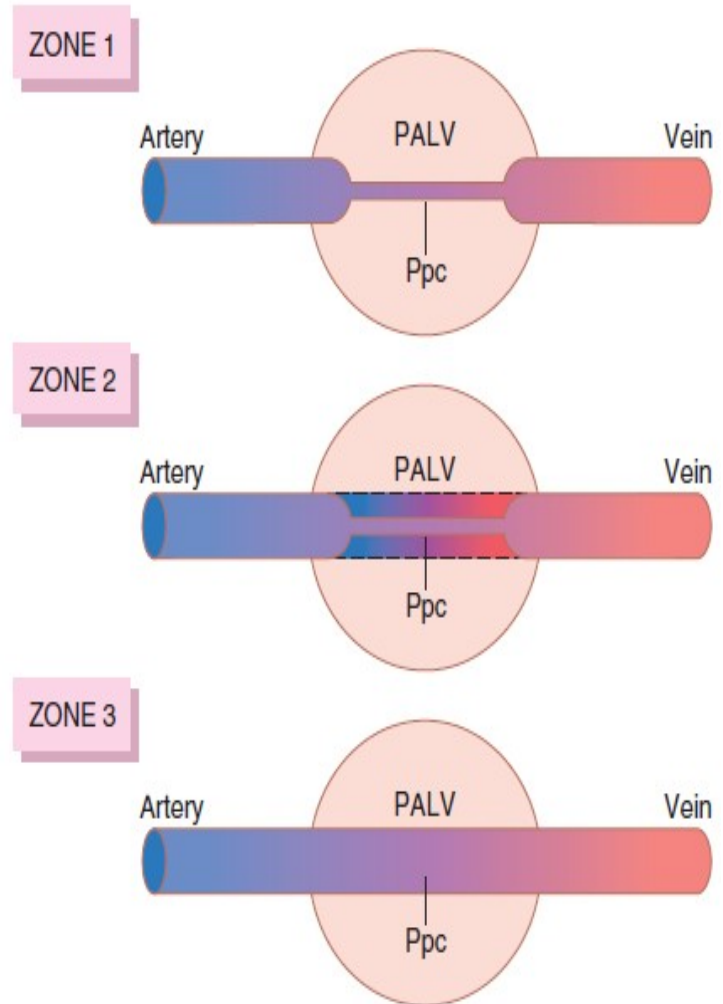
- ✓ alveolar air pressure (PALV) is greater than arterial pressure

ii. **Zone 2: Intermittent blood flow**

- ✓ systolic arterial pressure rises higher than alveolar air pressure, but diastolic arterial pressure falls below alveolar air pressure

iii. **Zone 3: continuous flow**

- ✓ arterial pressure and pulmonary capillary pressure (Ppc) remain greater than alveolar air pressure at all times.



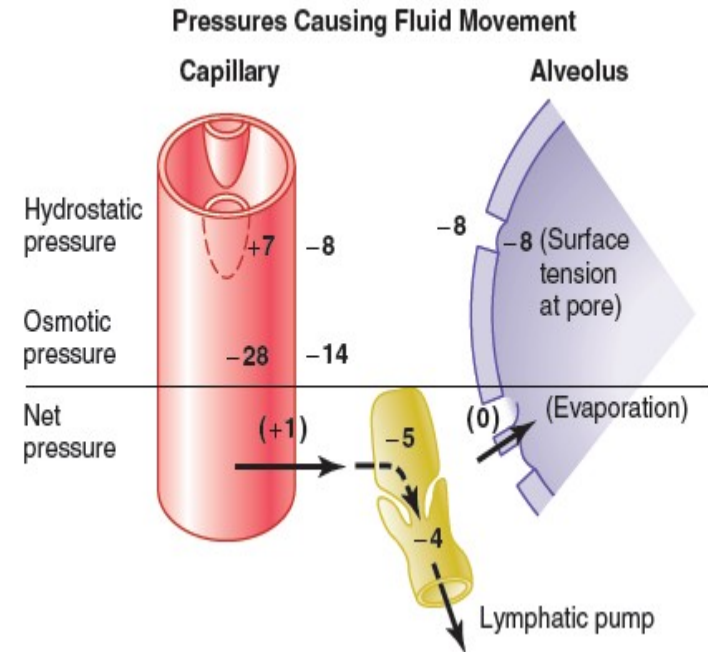
Pulmonary circulation

- ◆ Normally, the lungs have only zones 2 and 3 blood flow
 - ✓ Zone:2 (intermittent flow) in the apices and
 - ✓ Zone:3 (continuous flow) in all the lower areas.
- ◆ **Exercise** increases blood flow through all parts of the lungs (pulmonary vascular pressures rise enough during exercise to convert the *lung apices from a zone 2 pattern into a zone 3*)
- ◆ Zone :1 Blood Flow Occurs *Only Under Abnormal Conditions*
 - ✓ upright person Breathing against a positive air pressure so that the intra-alveolar air pressure is at least 10 mm Hg greater than normal but the pulmonary systolic blood pressure is normal (Zone :1 Apices)
 - ✓ upright person whose pulmonary systolic arterial pressure is exceedingly low, as might occur after severe blood loss

Pulmonary circulation

PULMONARY CAPILLARY DYNAMICS

- ◆ Q: What keeps the alveoli from filling with fluid under normal conditions?
- ✓ Pulmonary capillaries and the pulmonary lymphatic system normally maintain a slight *negative pressure in the interstitial space*-
Suction effect



2. Pulmonary Edema

- ◆ Any factor that increases fluid filtration out of the pulmonary capillaries or that impedes pulmonary lymphatic function
- ◆ The most common causes of pulmonary edema:-
 - i. Left-sided heart failure or mitral valve disease
 - ✓ Every time the left atrial pressure rises to high values, the pulmonary capillary pressure rises to a level 1 to 2 mm Hg greater than the left atrial pressure
 - ii. Damage to the pulmonary blood capillary membranes
 - ✓ By pneumonia or by breathing noxious substances such as chlorine gas or sulfur dioxide gas.

2. Pulmonary Edema

Reading Assignment

1. Pulmonary Edema Safety Factor

- ✓ When the pulmonary capillary pressure rises even slightly above the safety factor level, lethal pulmonary edema can occur within hours, or even within 20 to 30 minutes

3. Pleural Fluid

- ◆ When the lungs expand and contract during normal breathing, they slide back and forth within the pleural cavity.
- ◆ To facilitate this movement, a thin layer of *mucoïd fluid* lies between the parietal and visceral pleurae
- ◆ “Negative Pressure (−7 mm Hg)” in Pleural Fluid vs normal collapse tendency of the lungs about (−4 mm Hg)
- ◆ Pleural Effusion—Collection of Large Amounts of Free Fluid in the Pleural Space
 - ✓ blockage of lymphatic drainage from the pleural cavity
 - ✓ cardiac failure, which causes excessively high peripheral and pulmonary capillary pressures
 - ✓ greatly reduced plasma colloid osmotic pressure
 - ✓ infection or any other cause of inflammation of the surfaces of the pleural cavity

Thank you